

ENGINEERING SCIENCE (ESC)

2026-2027 DEGREE MAP

SEMESTER 1 13 CREDITS ENG 101 English Composition 3 CREDITS MAT 206 Precalculus¹ 4 CREDITS XXX xxx World Cultures and Global Issues² 3 CREDITS XXX xxx Individual and Society² 3 CREDITS	Create your Handshake Account and complete your profile. Begin exploring internships, jobs, and career paths that match your interests. Schedule an appointment with a career advisor at the Center for Career Development. Join a Student Club: Get involved on campus to start building your network and developing core skills. Consider joining the Science or other STEM-focused clubs. Activate your BMCC Co-Curricular Transcript (CCT). Joining a club and participating in a program will help you build your resume. Draft a Basic Resume: Start documenting your academic achievements, extracurriculars, and any work, internship, or volunteer experiences. Upload your resume to your Handshake profile. Research Baccalaureate Programs and Complete a Transfer Plan: Begin researching colleges and transfer requirements. Additional Department Specific Milestones - Identify research mentors and research programs available. - Begin building professional relationships with faculty to support mentorship, learn more about faculty area of expertise, and research opportunities. Start by attending office hours each semester.
SUMMER/ WINTER SESSION 4 CREDITS MAT 301 Analytic Geometry and Calculus I³ 4 CREDITS	
SEMESTER 2 18 CREDITS CHE 201 College Chemistry I⁴ 4 CREDITS ENG 201 Introduction to Literature 3 CREDITS MAT 302 Analytic Geometry and Calculus II⁵ 4 CREDITS PHY 215 University Physics I⁶ 4 CREDITS XXX xxx U.S. Experience in Its Diversity² 3 CREDITS	Create a LinkedIn Account: Establish a professional online presence. Contact the Center for Career Development for support. Attend Professional Development Events and Workshops, including the BARS Research Symposium, career days, and job and internship fairs. Draft a Basic Cover Letter: Practice writing a cover letter that you can use to apply to internships and fellowships. Review job postings and Career Coach to identify the skills employers want, and highlight them in your cover letter. Upload your cover letter to your Handshake profile. Attend Transfer Events: Receive assistance with the transfer process, admissions requirements, transfer credit evaluation, and more. Additional Department Specific Milestones - Identify competitive summer research programs (SURPs and REUs) of interest, like at Northrop Grumman and other companies. - Strengthen a faculty mentoring relationship.

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SEMESTER 3 18 CREDITS CHE 202 College Chemistry II⁷ 4 CREDITS ESC 111 Elements of Engineering Design⁸ 1 CREDIT MAT 303 Analytic Geometry and Calculus III⁹ 4 CREDITS PHY 225 University Physics II¹⁰ 4 CREDITS XXX xxx Curriculum Elective¹¹ 5 CREDITS	Find a Mentor to Check in With Periodically: Establish a relationship with someone who can provide guidance and advice related to your career goals. Start Your Internship Search: Begin looking for and applying for CUNY-wide, college-specific, and external internship opportunities relevant to your major. Research and Apply to Baccalaureate Programs: Apply to colleges that offer a BA/BS degree in your field. Recommended milestones: Join a Professional Organization or Pursue Leadership Roles: Deepen your engagement in your field of study by joining a professional organization (e.g., National Society of Professional Engineers, The Honor Society). Develop a Network with LinkedIn: Actively connect with professionals and alumni in your field. Update Your Resume: Incorporate any new experiences or skills acquired since your last update. Identify Skills You Need to Build Outside the Classroom: Use job descriptions and insights from professionals to identify in-demand skills in your career field. Then, explore tools and resources to develop those skills (e.g. Coursera, CUNY Upskilling, LinkedIn Learning). Additional Department Specific Milestones - Participate in faculty-guided research program and/or honors project with formal presentation. - Attend a local or regional engineering or related STEM conference.
SEMESTER 4 16 CREDITS ESC 113 Computer Aided Analysis for Engineering¹² 2 CREDITS MAT 501 Ordinary Differential Equations¹³ 3 CREDITS SCI 120 Computer Methods in Science 4 CREDITS SPE 100 Fundamentals of Public Speaking¹⁴ 3 CREDITS XXX xxx Curriculum Elective¹¹ 4 CREDITS	Complete Graduation Survey in CUNYfirst Checklist Accept Your Offer to Enroll in a Baccalaureate Program: Accept your offer through CUNYfirst, if applicable and complete any supplemental materials. Recommended milestones: Attend Networking Events and Career Fairs: Attend networking events and career fairs to meet employers, build connections, and learn about opportunities. Final Interview Preparation: Schedule interview practice sessions with the Center for Career Development or use the resources on Handshake to refine your interview skills. Additional Department Specific Milestones - Present research at the BARS symposium. - Apply to SURPs and REUs.
TOTAL: 69 CREDITS¹⁵	
Footnotes ¹ Students are required to take MAT 206. MAT 206 will satisfy the Mathematical & Quantitative Requirement. Please speak to your faculty/ academic advisor about your appropriate math pathway. ² Please consult with an academic or faculty advisor. ³ MAT 206 must be passed in order to take MAT 301. ⁴ Students must be eligible to take MAT 206 in order to take CHE 201. ⁵ MAT 301 must be passed in order to take MAT 302. ⁶ PHY 215 has a corequisite of MAT 302. ⁷ CHE 201 must be passed in order to take CHE 202. ⁸ MAT 206 and CHE 201 must be passed in order to take ESC 111.	⁹ MAT 302 must be passed in order to take MAT 303. ¹⁰ PHY 215 must be passed in order to take PHY 225. ¹¹ A total of 9 credits is required. Choose from CHE 230, CHE 240, ESC 130, ESC 131, ESC 201, ESC 202, ESC 211, ESC 221, ESC 223, GLY 210, MAT 315, or PHY 240. ¹² The following courses must either be taken in the same semester as ESC 113 or must have already been passed prior to taking ESC 113- CHE 201, MAT 206 and PHY 215 or departmental permission will be required. ¹³ MAT 302 must be passed in order to take MAT 501. ¹⁴ SPE 102 is an option for non-native speakers of English. ¹⁵ A Writing Intensive course is needed to graduate.